9. Simulation and Performance Evaluation of OFDMA Resource-Block Allocation for Multi-User Communication.

Objectives of This Experiment

1. To represent a set of orthogonal OFDMA resource blocks and multiple users in MATLAB.

2. To allocate resource blocks using traffic-demand and channel-quality-based allocation rules.

3. To compare user throughput, resource utilization and fairness under different allocation strategies.

4. To explain how orthogonal resource allocation limits inter-user interference in multi-user communication.

Expected Outcome from This Experiment

Students shall be able to:

• Develop a MATLAB model of an OFDMA downlink containing multiple users and resource blocks.

• Implement round-robin, channel-quality-based and traffic-aware resource-block allocation.

• Calculate and compare individual-user throughput, total system throughput, resource utilization and fairness.

• Interpret the role of orthogonal resource allocation in improving multi-user communication performance.

Software Required: MATLAB (with Communications and Signal Processing Toolboxes if needed).

MATLAB Program Script

```
clc;
clear;
```

```
close all;
%% Input parameters
rng(7); % Repeatable random results
K = 4; % Number of users
NRB = 24; % Number of resource blocks
BRB = 180e3; % Bandwidth per RB in Hz
Tslot = 1e-3; % Scheduling interval in seconds
etaMax = 6; % Maximum spectral efficiency, bit/s/Hz
avgSNRdB = [5 10 15 8]; % Average user SNR in dB
trafficMbps = [8 12 5 10]; % Offered traffic in Mbit/s
% Bits requested during one scheduling interval
demandBits = trafficMbps * 1e6 * Tslot;
%% Generate channel quality for each user and RB
% Independent SNR variation around each user's average SNR
snrVariation = 4 * randn(K, NRB);
snrDB = avgSNRdB(:) + snrVariation;
snrLinear = 10.^(snrDB / 10);
% Shannon-based spectral efficiency with an upper limit
spectralEfficiency = min(log2(1 + snrLinear), etaMax);
% Achievable bits on each RB during one scheduling interval
rateBps = BRB * spectralEfficiency;
bitsPerRB = rateBps * Tslot;
strategyNames = {'Round Robin', 'Maximum CQI', 'Traffic Aware'};
allocations = zeros(K, NRB, 3);
%% Strategy 1: Round-robin allocation
for n = 1:NRB
user = mod(n - 1, K) + 1;
allocations(user, n, 1) = 1;
```

```
end
%% Strategy 2: Maximum-CQI allocation
for n = 1:NRB
[~, user] = max(snrDB(:, n));
allocations(user, n, 2) = 1;
end

%% Strategy 3: Traffic-aware channel-quality allocation
servedBits = zeros(K, 1);
for n = 1:NRB
remainingBits = max(demandBits(:) - servedBits, 0);
if all(remainingBits <= 0)
break;
end
normalizedDemand = remainingBits ./ max(demandBits(:), eps);
metric = bitsPerRB(:, n) .* normalizedDemand;
% Prevent allocation to a user whose demand has been satisfied
metric(remainingBits <= 0) = -inf;
[~, user] = max(metric);
allocations(user, n, 3) = 1;
deliveredBits = min(bitsPerRB(user, n), remainingBits(user));
servedBits(user) = servedBits(user) + deliveredBits;
end
%% Performance calculations
userThroughputMbps = zeros(K, 3);
totalThroughputMbps = zeros(1, 3);
jainFairness = zeros(1, 3);
utilization = zeros(1, 3);
```

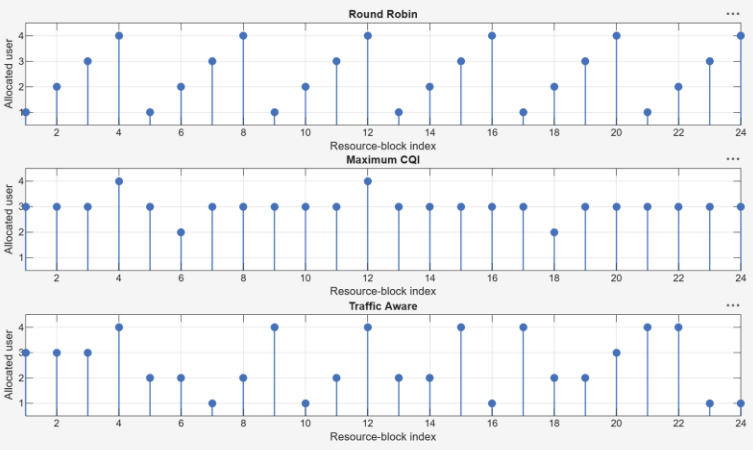
```
unservedTrafficMbps = zeros(K, 3);
rbPerUser = zeros(K, 3);
for s = 1:3
allocation = allocations(:, :, s);
deliveredBits = sum(allocation .* bitsPerRB, 2);
% Throughput is restricted to requested traffic
usefulBits = min(deliveredBits, demandBits(:));
userThroughputMbps(:, s) = usefulBits / Tslot / 1e6;
totalThroughputMbps(s) = sum(userThroughputMbps(:, s));
throughput = userThroughputMbps(:, s);
jainFairness(s) = sum(throughput)^2 / (K * sum(throughput.^2) + eps);
allocatedRBs = sum(allocation(:));
utilization(s) = allocatedRBs / NRB * 100;
unservedTrafficMbps(:, s) = max(trafficMbps(:) - throughput, 0);
rbPerUser(:, s) = sum(allocation, 2);
end
%% Display results

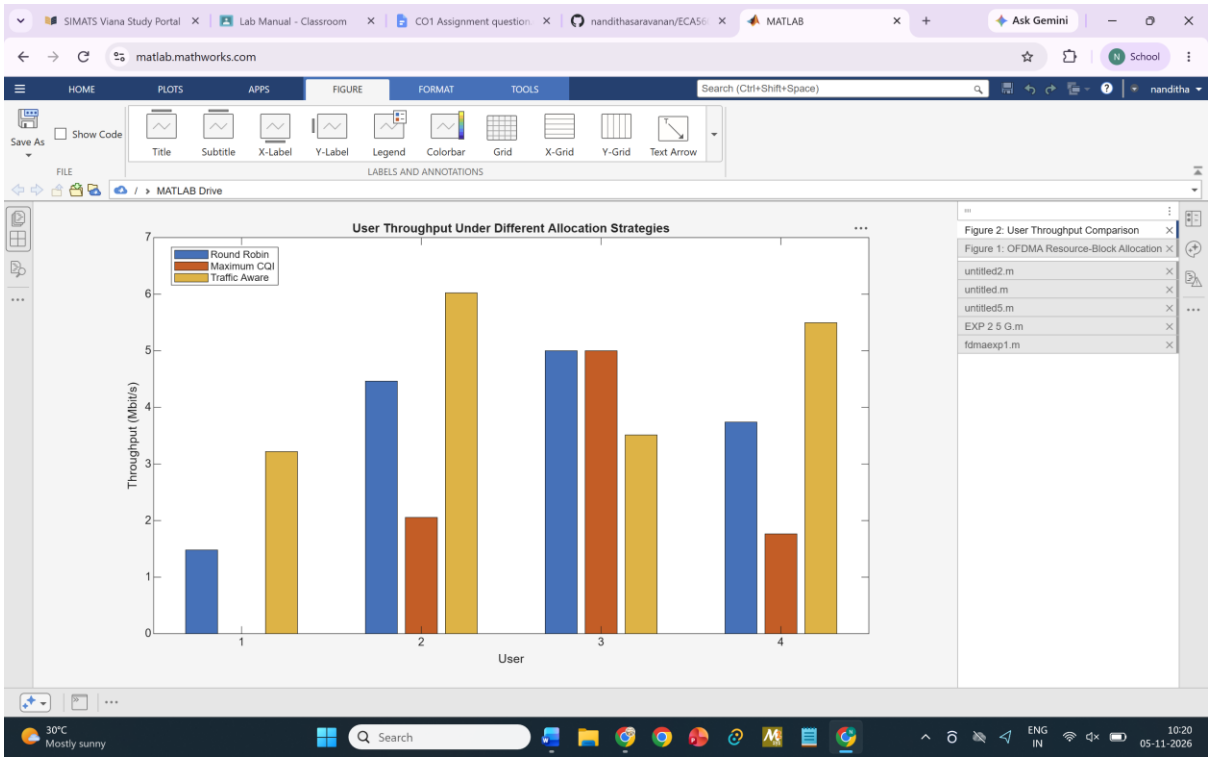
fprintf('\nOFDMA RESOURCE-BLOCK ALLOCATION RESULTS\n');
fprintf('=======================================\n');
for s = 1:3
fprintf('\n%s\n', strategyNames{s});
fprintf('User\tRBs\tThroughput (Mbit/s)\tUnserved (Mbit/s)\n');
for k = 1:K
fprintf('%d\t%d\t%8.3f\t\t%8.3f\n', ...
k, rbPerUser(k, s), userThroughputMbps(k, s), unservedTrafficMbps(k, s));
end
fprintf('Total throughput: %.3f Mbit/s\n', totalThroughputMbps(s));
```

```
fprintf('Jain fairness : %.4f\n', jainFairness(s));
fprintf('RB utilization : %.2f %%%%\n', utilization(s));
end
%% Allocation maps plotting
figure('Name', 'OFDMA Resource-Block Allocation');
for s = 1:3
subplot(3, 1, s);
[userIndex, rbIndex] = find(allocations(:, :, s));
stem(rbIndex, userIndex, 'filled', 'LineWidth', 1.1);
grid on;
xlim([1 NRB]); ylim([0.5 K + 0.5]); yticks(1:K);
xlabel('Resource-block index'); ylabel('Allocated user');
title(strategyNames{s});
end
%% Throughput comparison plotting
figure('Name', 'User Throughput Comparison');
bar(userThroughputMbps);
xlabel('User'); ylabel('Throughput (Mbit/s)');
title('User Throughput Under Different Allocation Strategies');
legend(strategyNames, 'Location', 'best');
```

Output:

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```




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Legend
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Y-Grid
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LABELS AND ANNOTATIONS
MATLAB Drive
User Throughput Under Different Allocation Strategies
Round Robin
Maximum CQI
Traffic Aware
Throughput (Mbit/s)
User
0
1
2
3
4
5
6
7
1
2
3
4
Figure 2: User Throughput Comparison
Figure 1: OFDMA Resource-Block Allocation
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30°C
Mostly sunny
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